

Day 10_100 Days Challenge – Cyber Security & Ethical Hacking

Proxies & VPNs: Identity Obfuscation, Privacy, and Attribution Warfare

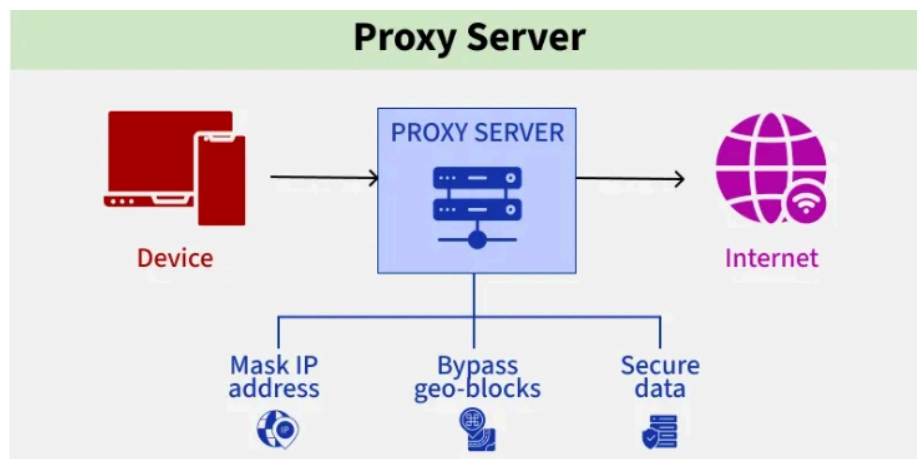
♦ Core Cyber Insight

- Proxies and VPNs are **neutral technologies**
 - Their impact depends on **intent and usage**
 - Attackers use them for **anonymity and evasion**
 - Investigators use them for **privacy, containment, and lawful analysis**
 - Cybersecurity requires understanding **both sides of attribution**
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♦ Definitions:

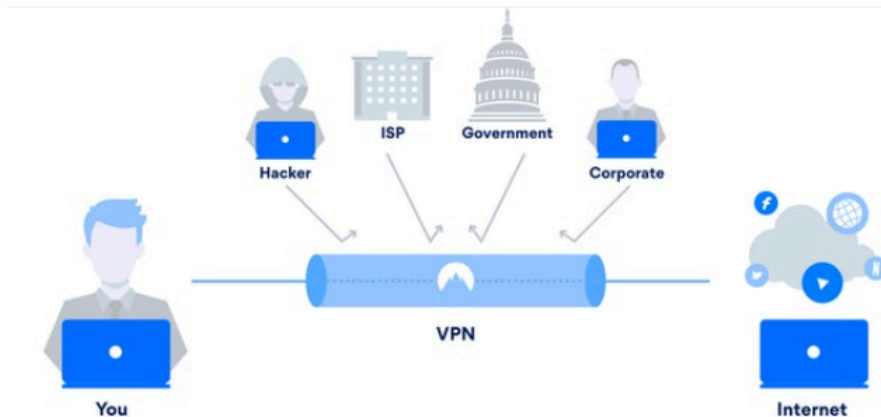
• Proxy Server

- An intermediary that forwards traffic between a client and a target
- Can modify headers, IP visibility, and routing
- Often application-layer specific (HTTP/SOCKS)



- **Virtual Private Network (VPN)**

- An encrypted tunnel between user and VPN endpoint
- Masks original IP and secures traffic in transit
- Operates at the network layer, affecting all traffic



- ◆ **Why Attackers Use Proxies & VPNs (Threat Actor Perspective)**

- To **hide real IP address and location**
- To **bypass IP-based rate limiting and bans**
- To **rotate identities during reconnaissance**
- To **blend malicious traffic with normal user traffic**
- To complicate **attribution and incident response**

⚠ Advanced attackers rarely rely on a *single* layer
They use **multi-hop chains** (proxy → proxy → VPN → target)

◆ Proxy Types Commonly Observed in Attacks

- **Open/Compromised Proxies**

- Misconfigured servers abused without owner knowledge

- **Residential Proxies**

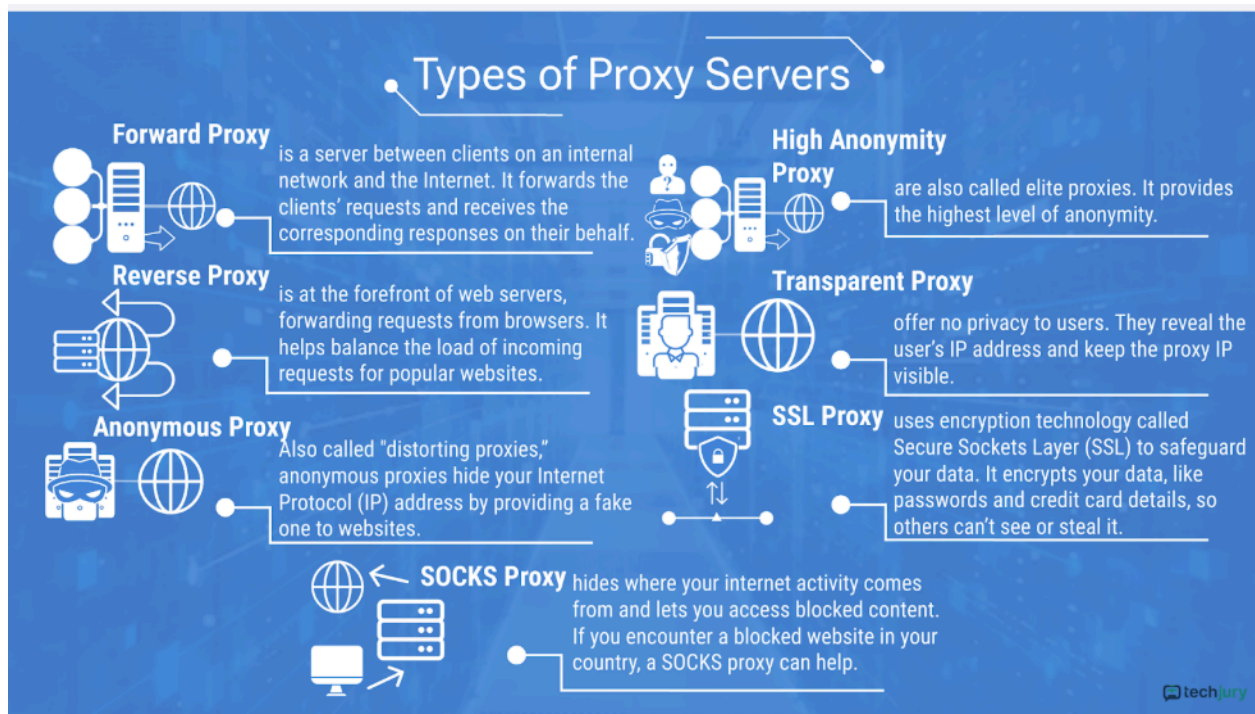
- Traffic appears to come from real home users
- Extremely difficult to block

- **Mobile Proxies**

- Use carrier-grade IP ranges
- High trust score, low detection rate

- **SOCKS Proxies**

- Used for flexibility across protocols



♦ VPN Usage Patterns in Cyber Operations

- **Attackers:**

- Use VPNs as **first-layer obfuscation**
- Chain VPNs with proxies or compromised hosts

- **Investigators & professionals:**

- **Use VPNs to:**
 - Protect identity during OSINT
 - Avoid tipping off threat actors
 - Safely analyze malicious infrastructure

♦ Real-World Cyber Security Example

- **During a phishing or credential-stuffing campaign:**

- Login attempts originate from:
 - Multiple countries
 - Residential and mobile IP ranges
- This prevents:
 - Geo-blocking
 - Simple IP blacklisting

- **SOC teams detect:**

- Behavioral anomalies
- Impossible travel patterns
- ASN reputation inconsistencies

◆ Impact on Attribution & Investigations

- **VPNs and proxies:**

- Delay attacker identification
- Increase false positives
- Obscure jurisdiction

- **Investigators counter this using:**

- Traffic correlation
- Timing analysis
- Endpoint and behavioral telemetry
- Legal cooperation with providers (where applicable)

Day 10 Master-Level Conclusion

- Proxies and VPNs are **core tools in modern cyber operations**
- Attackers use them to **hide**
- Defenders must learn to **see through them**
- True cybersecurity skill lies in **understanding intent, patterns, and context**
- **Attribution is a war of layers—not a single IP lookup**